

Polish Math Olympiad 2000 — P5

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Specifically from the first round of the 52nd Polish MO.

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Show that for all integer $n \geq 2$ and for all prime numbers p the number

$$n^{p^p} + p^p$$

is composite.

Solution

We need the following fact.

Claim 1: For $a, b \in \mathbb{N}$ and odd $k \in \mathbb{N}$, the following is true $(a + b) \mid (a^k + b^k)$.

Proof: We can use factor theorem. If we let $f(a) = a^k + b^k$, then we notice that $f(-b) = (-b)^k + b^k = 0$ since k is odd, and this means that $f(a)$ has a factor of the form $(a + b)$. $\square$

Now observe that, for odd primes p

$$n^{p^p} + p^p = \left(n^{p^{p-1}}\right)^p + (p)^p$$

but due to claim 1, this number is divisible by $n^{p^{p-1}} + p$. Now since prime numbers are only divisible by 1 and themselves the number $n^{p^p} + p^p$ cannot be prime.

Then when $p = 2$ which is the only even prime it factors via Sophie Germain's identity

$$n^4 + 4 = (n^2 - 2n + 2)(n^2 + 2n + 2)$$

and for $n \geq 2$ it is easy to show that the factor $1 < n^2 + 2n + 2$ is strictly less than $n^4 + 4$. Thus $n^4 + 4$ has a factor other than 1 and itself and so isn't prime. $\blacksquare$

I thought the other problems were interesting also, so here are some selected solves to other problems on the same exam.

Polish Math Olympiad 2000 — P3

Find all positive integers $n \geq 2$, such that the inequality

$$x_1 x_2 + x_2 x_3 + \cdots + x_{n-1} x_n \leq \frac{n-1}{n} (x_1^2 + x_2^2 + \cdots + x_n^2)$$

is satisfied for all positive real numbers $x_1, x_2, \dots, x_n$.

Solution

When $n = 2$ we have

$$x_1 x_2 \leq \frac{1}{2} (x_1^2 + x_2^2)$$

which is true by a direct application of AM-GM. Now we will show that for any $n > 2$ the given inequality always has a counter-example.

Use $x_1 = x_2 = \cdots = x_{n-1}$ and $x_n = M$. We have

$$\begin{aligned} (n-2) + M &\leq \frac{n-1}{n} [(n-1) + M^2] \\ n(n-2) + nM &\leq (n-1) [(n-1) + M^2] \\ 0 &\leq (n-1)M^2 - nM + 1 \end{aligned}$$

We can check that this quadratic in M has the solutions $1, 1/(n-1)$. Thus if we pick

$$M \in \left(\frac{1}{n-1}, 1 \right)$$

then the inequality fails, and so indeed for every $n > 2$ we have a counter-example. ■

Polish Math Olympiad 2000 — P6

The integers a, b, x, y satisfy the equality

$$a + b\sqrt{2001} = (x + y\sqrt{2001})^{2000}.$$

Prove that $a \geq 44b$.

Solution

Look at the expansion of the RHS.

$$\sum_{r=0}^{2000} \binom{2000}{r} (y\sqrt{2001})^r x^{2000-r}$$

If I replaced y with $-y$ then the only terms that would be affected would be the terms that have a factor of $\sqrt{2001}$ since they occur when r is odd, whilst terms that occur when r is even aren't affected since $(-y)^2 = y^2$. More precisely if I replace y with $-y$ then all of the terms that have a factor of $\sqrt{2001}$ have their signs changed. That means

$$a - b\sqrt{2001} = (x - y\sqrt{2001})^{2000} \geq 0$$

and so

$$a \geq b\sqrt{2001}.$$

But since a is an integer,

$$a \geq b \left\lceil \sqrt{2001} \right\rceil = 44b. \blacksquare$$